Health Happens Somewhere: What Might GPS Data Tell Us?

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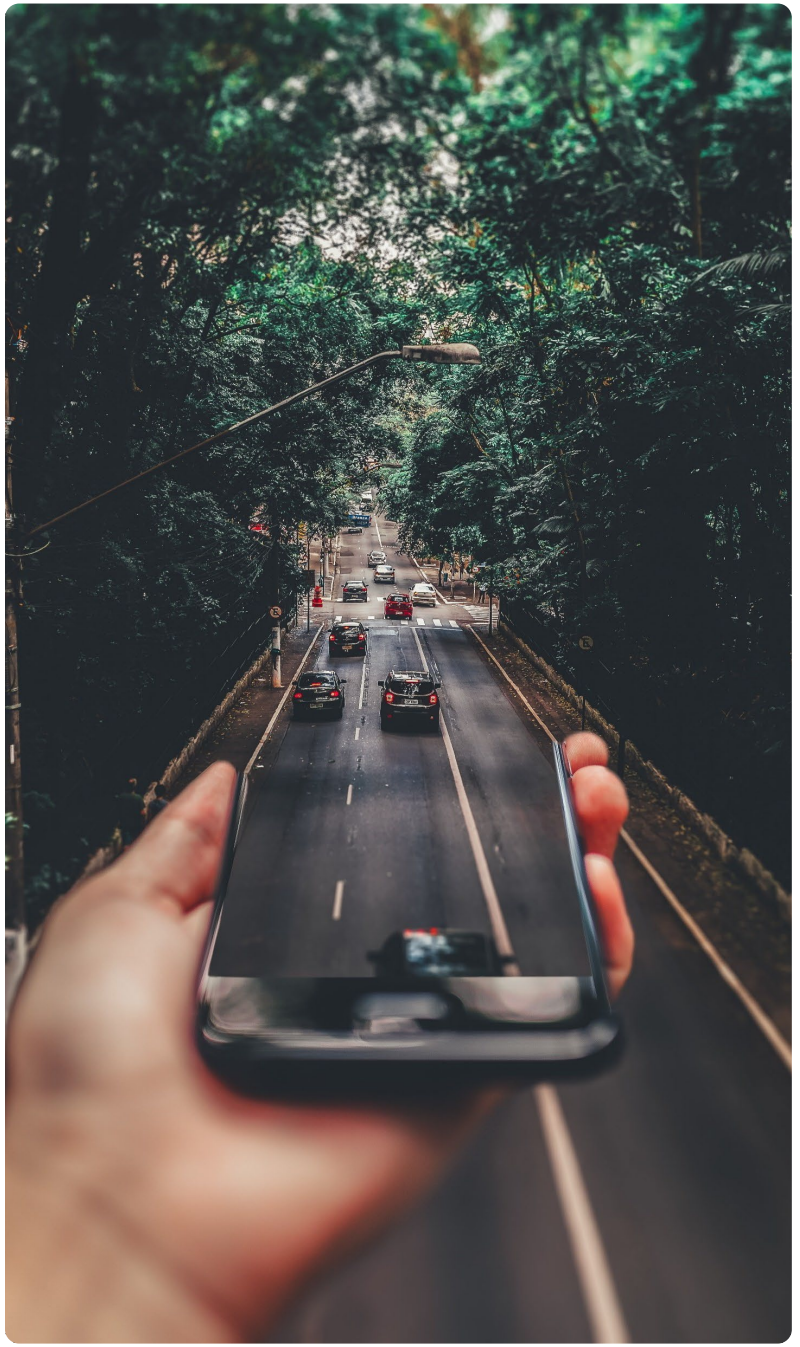
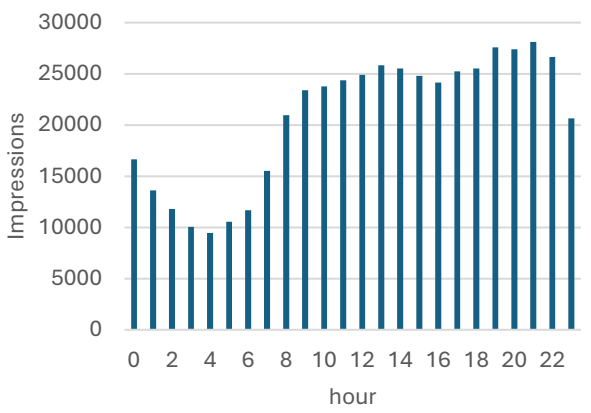


A day in Glasgow

Mean hourly user mobility for a Thursday in June 2020

This GIF is a visual representation of the mean hourly location [jittered] for de-identified mobile phone users in the raw data. This relates to the volume of data in the chart and is shown to illustrate the granularity of the data but jittered for privacy purposes.

As the hours increase throughout the day, we begin to users move in the data and the city come to life as we might expect.



- De-identified mobile device (jittered)

Using mobile phone app data to explore use of public space

Developing and exploring methods to understand **human-nature interactions** in urban areas using new forms of big data

UKRI Economic and Social Research Council

Capturing engagement numbers at ticketed and non-ticketed **cultural and sports events and spaces**

Department for Digital, Culture Media & Sport

Key partners

Glasgow City Council

greenspace scotland

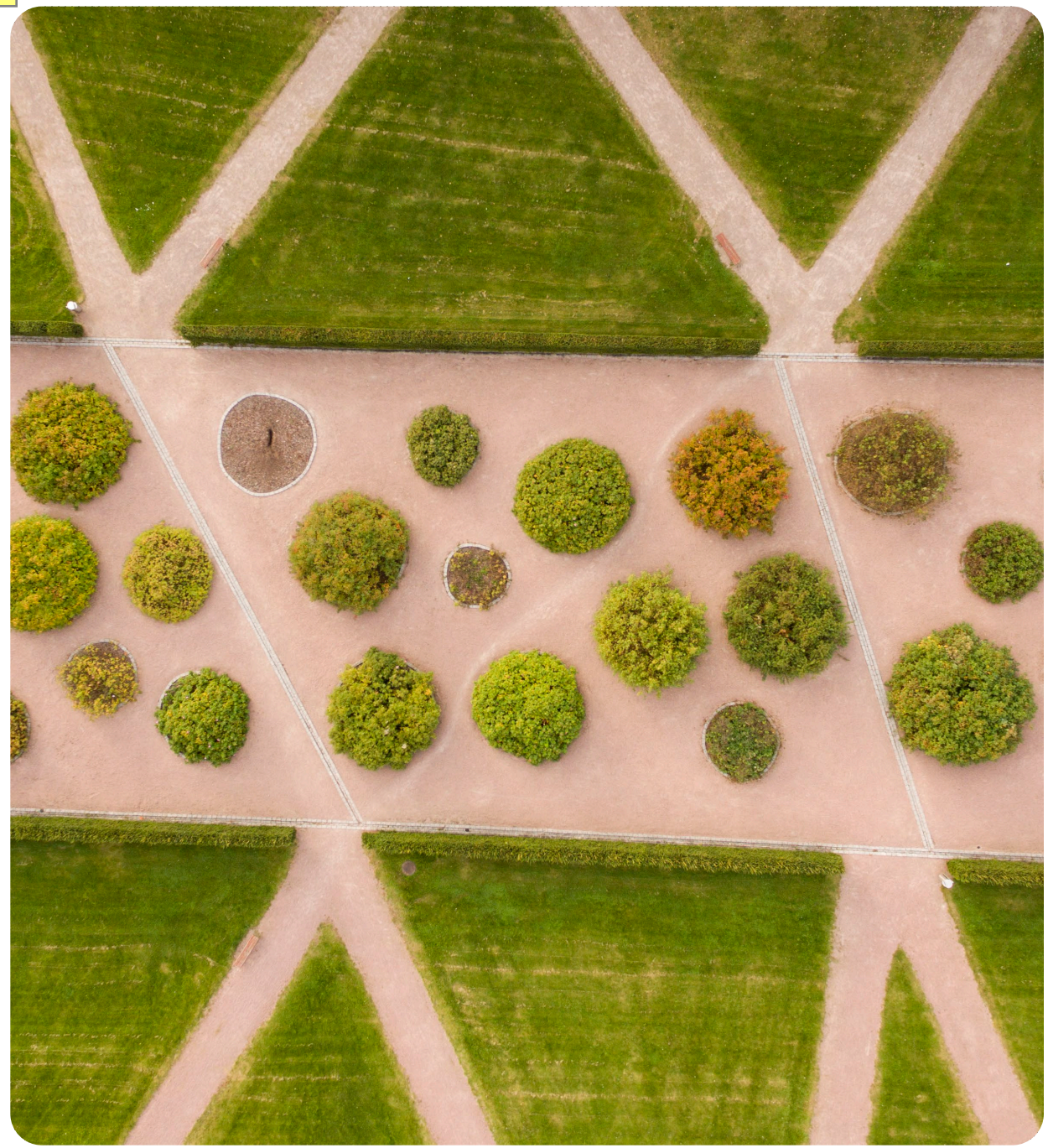
ParkPower

Transforming urban spaces for green energy

University of Glasgow

verian

faculty



Exploring patterns of park use from GPS data from mobile apps

The vast quantity of this data and its granularity offer new ways to understand how parks are being used in space and where visitors originate. Enriching the data with home area estimates also allows to explore patterns of use for different groups.

Guiding research questions:

Can the data show where visitors come from?

Can the data help us understand spatial patterns of use?

Can the data help us understand temporal patterns for different groups of people?

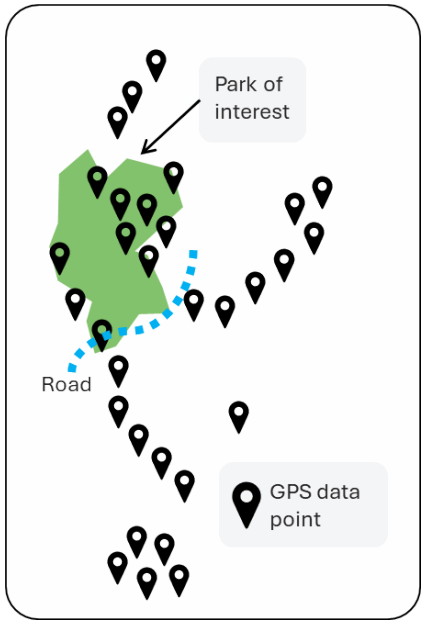
Glasgow city-region: Public parks and gardens

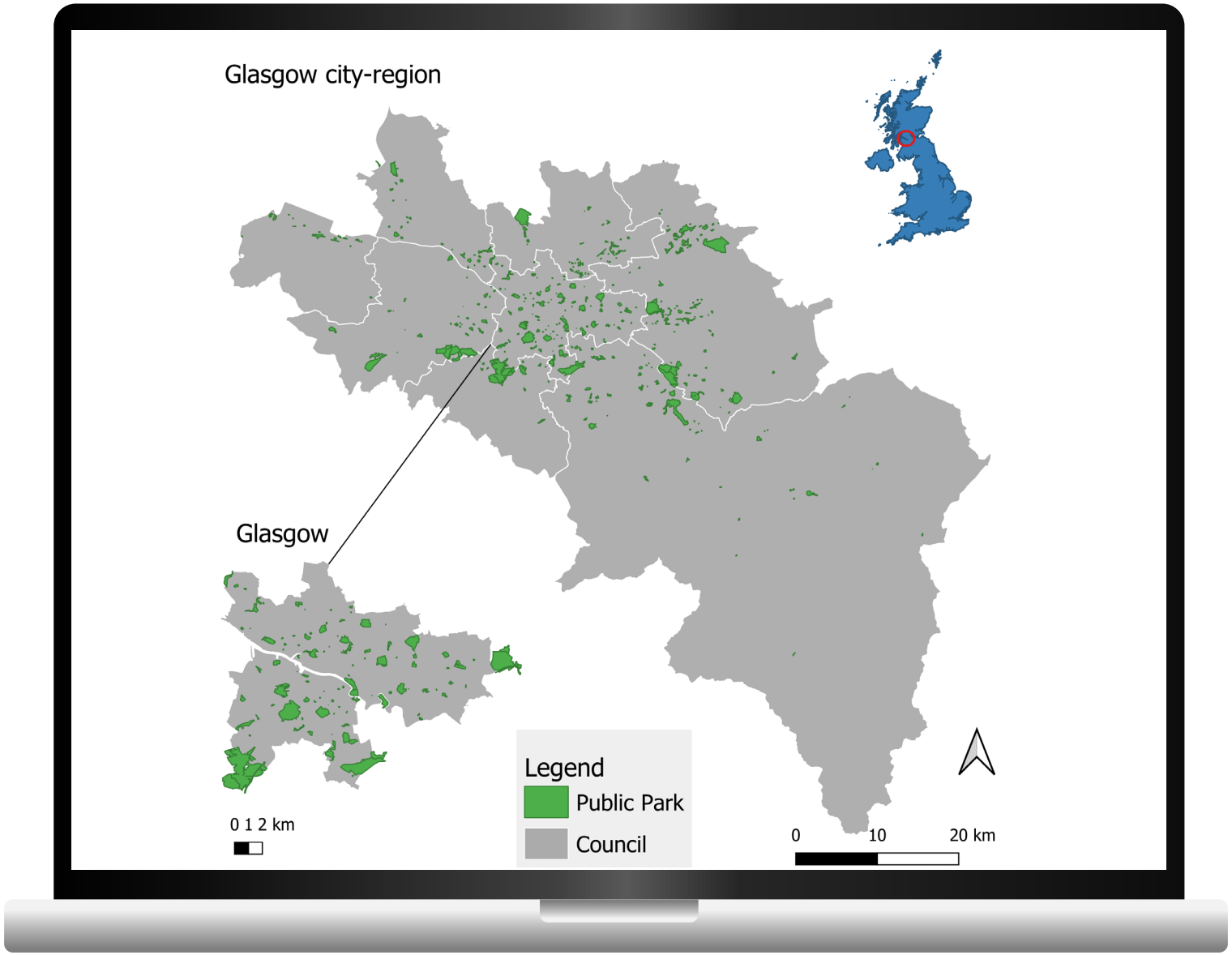
[Ordinance Survey open greenspace data]

GLASGOW CITY-REGION

Public parks and gardens from Ordinance Survey Open Greenspace data.

392 parks





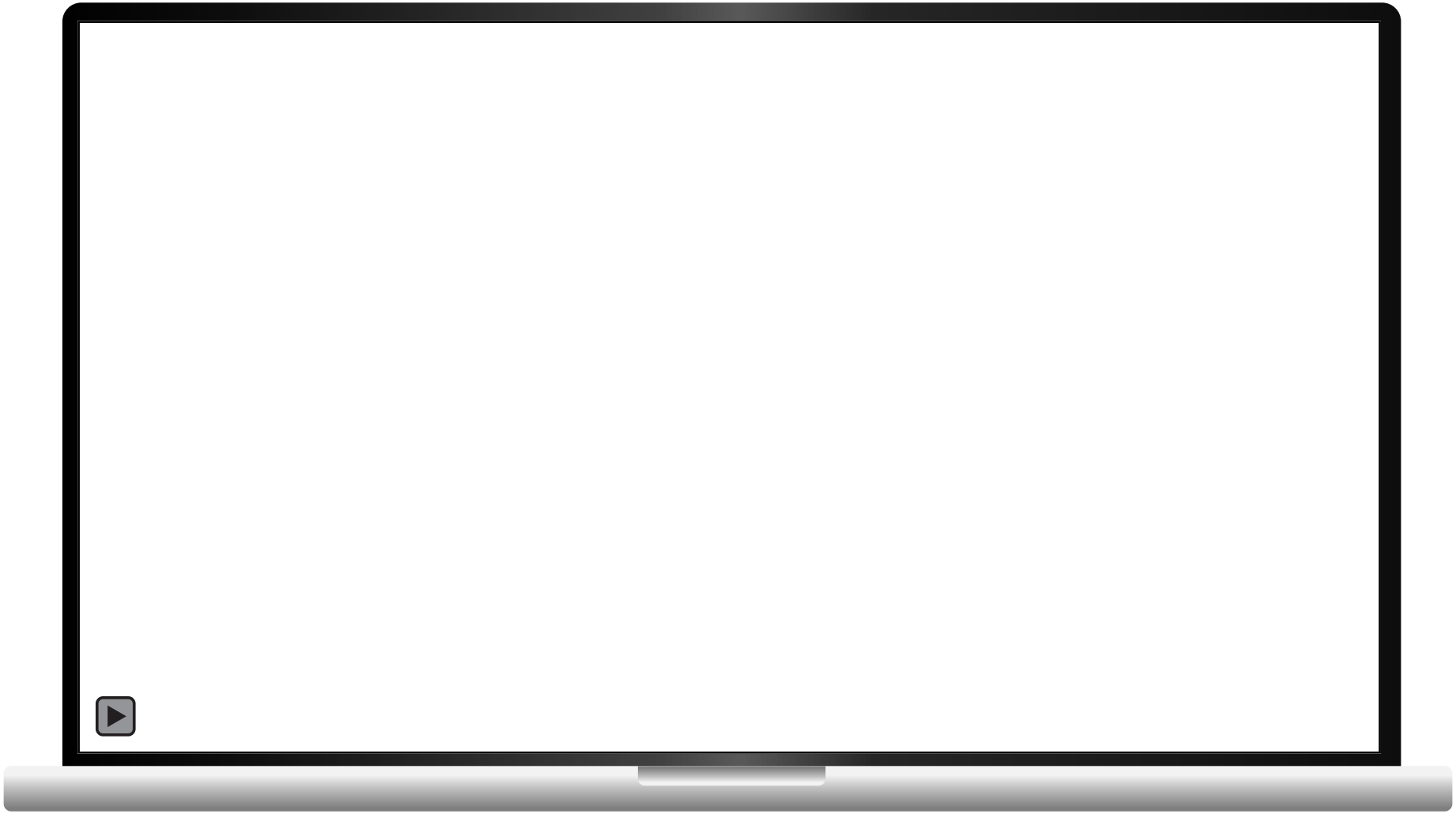
Spatial patterns of park use at scale

Using the accuracy and resolution of the mobile data to map patterns at park level

SUMMARY

Given the granularity of the data, we can map and explore patterns of use for parks in a way which is not possible with more traditional methods in social science.

Aggregating this is an important ethical consideration. We use UberH3.



Where do people visit parks from?

The enriched user data allows to explore the catchment of parks and where visits originate

SUMMARY

We can also explore the catchment or reach of individual parks.

This visualisation shows an example of the reach for a few parks in Glasgow.

Exploring park-specific use with GPS data from apps

Large and popular country park outside Glasgow

STRATHCLYDE COUNTRY PARK

Strathclyde Country Park is a large country park in the Clyde Valley, built around Strathclyde Loch. It's a recreation site offering walking and cycling routes, a 3.5-mile loch circuit, and extensive watersports including rowing, sailing and kayaking.

Location: North Lanarkshire

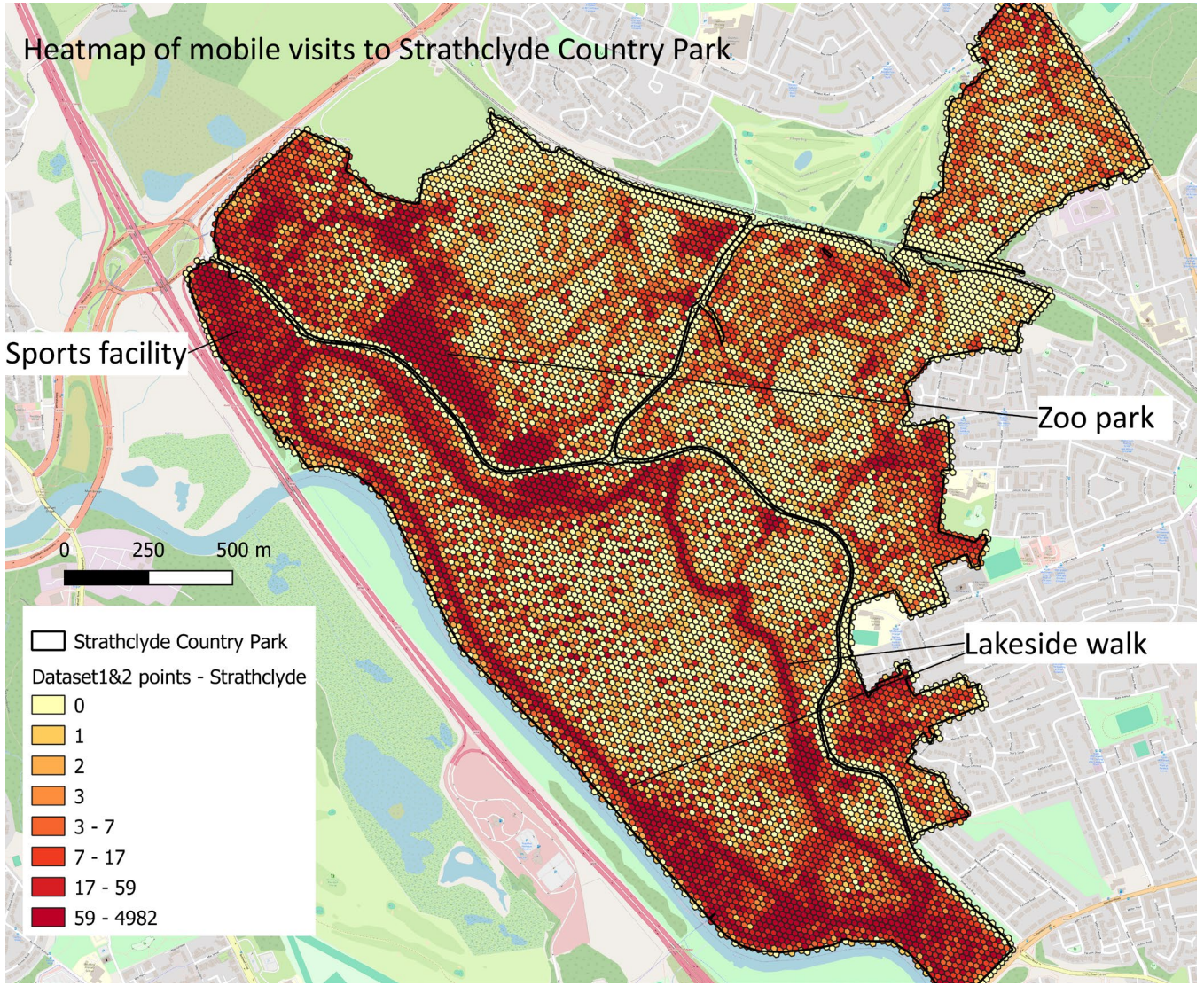
Spatial patterns of visitors in Strathclyde Country Park

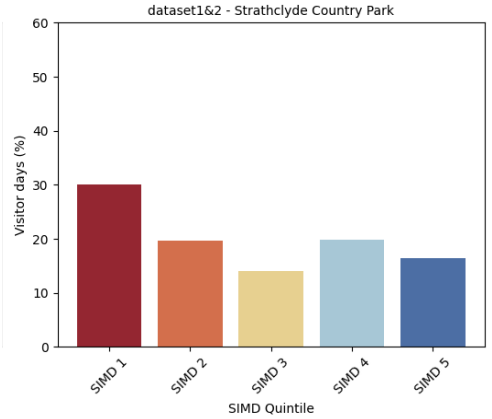
Using the accuracy and resolution of the mobile data to map patterns at park level

SUMMARY

Given the granularity of the data, we are able to visualise and explore patterns of park visitors at a scale unlike more traditional forms of data.

This shows results for both datasets combined





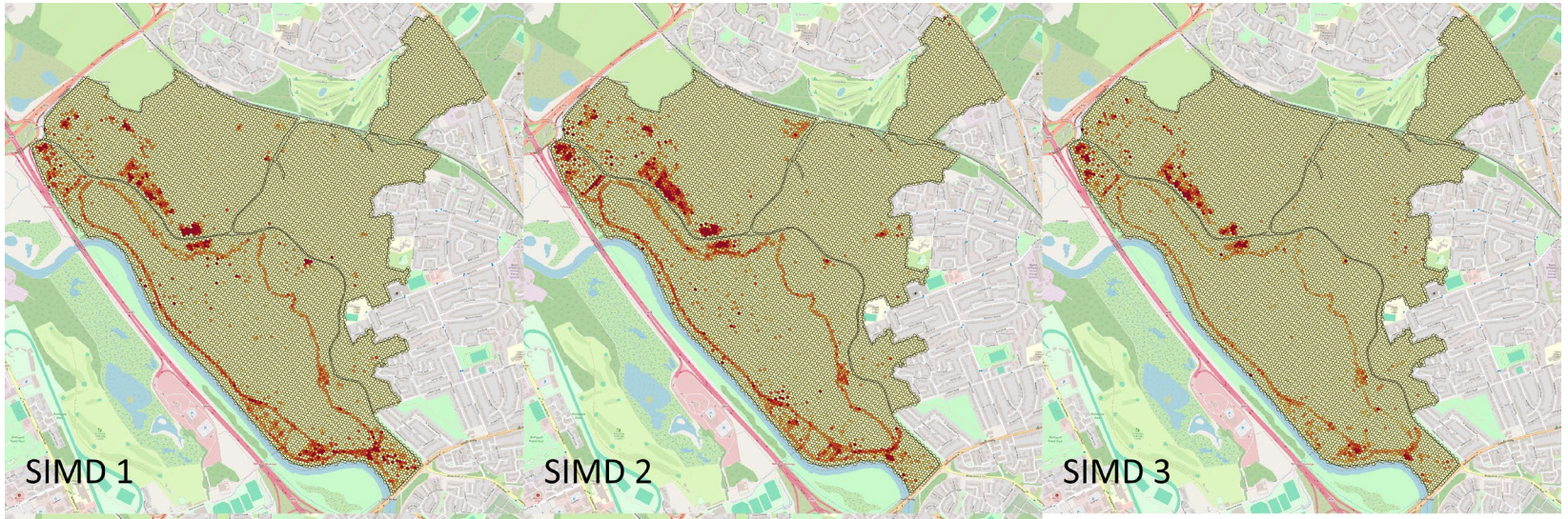
CHALLENGE

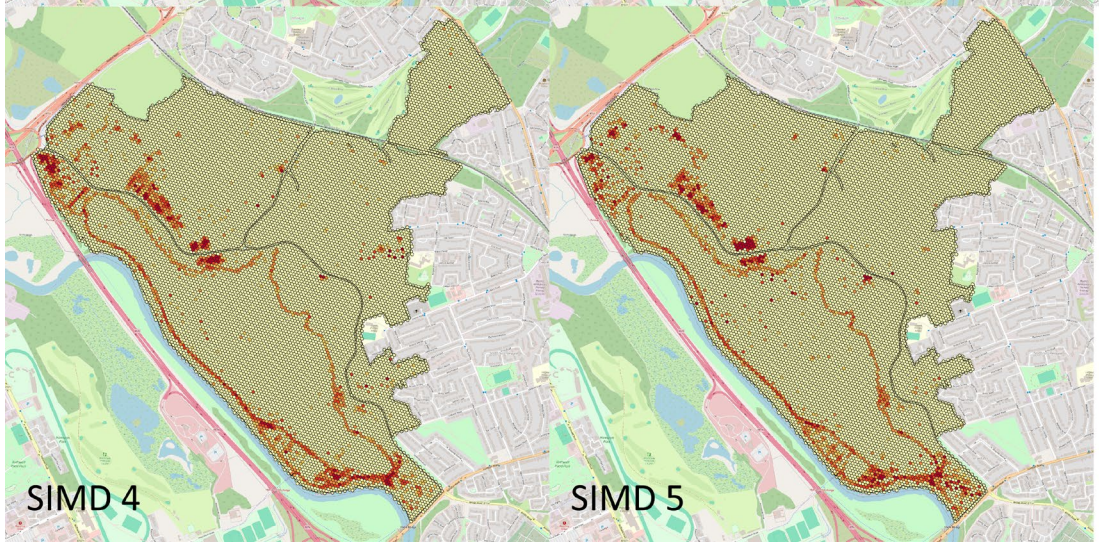
Aggregating the raw mobile phone app data is an important ethical consideration.

Here we use UberH3 polygons at a resolution which allows patterns to be explored but masks the points-level data.

We can explore patterns of use by visitor demographic

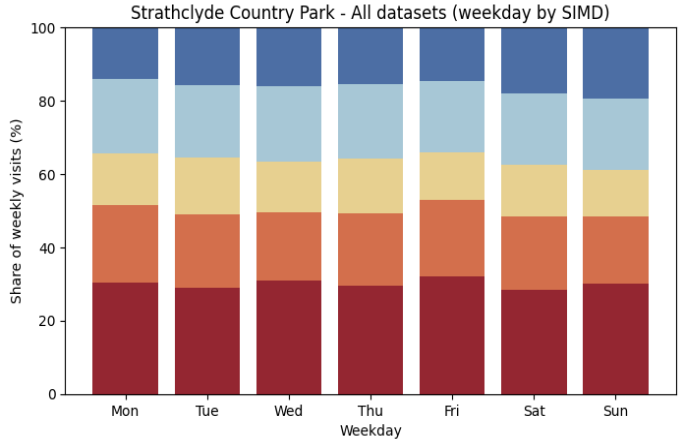
By enriching the data, we can map patterns at park level for different SIMD quintile groups

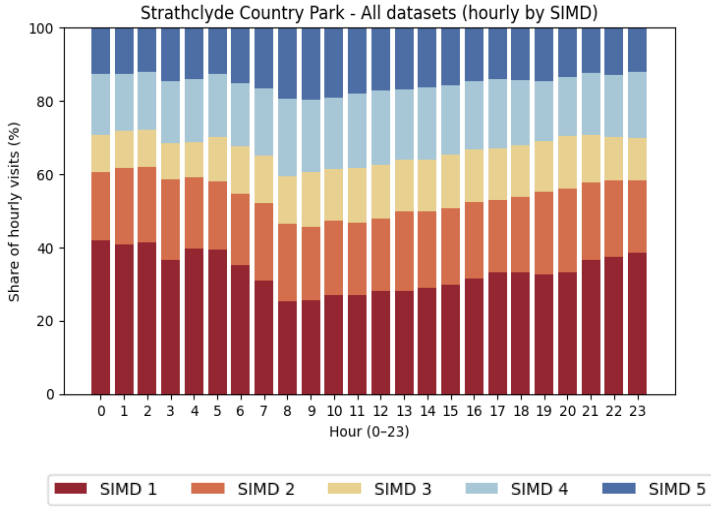




Dataset 1&2 points - Strathclyde

- 0 - 10
- 10 - 20
- 20 - 50
- 50 - 100
- >100





Key Challenges in Mobile Phone Data in Social Research

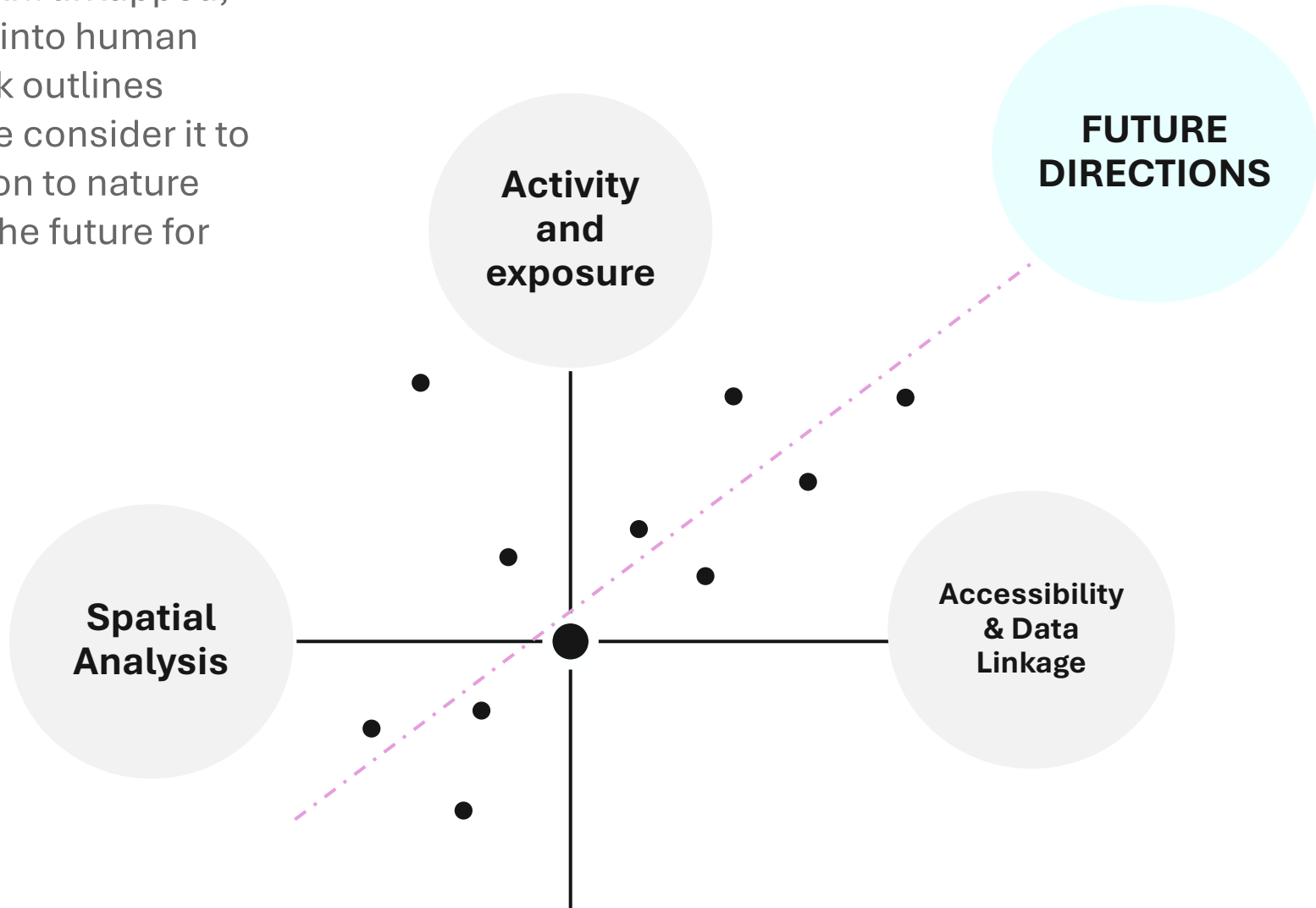
Contextualising the Main Issues

Mobile phone data offers valuable insights into human behaviour in nature and other public spaces, but its use also raises important challenges. This framework outlines four core areas to consider. The were learned through the context of application to nature but are relevant to social science research in general.

Key Opportunities in Mobile Phone Data Social Research

Contextualising the Main Opportunities

Mobile phone data offers many opportunities which are still untapped, offering valuable insights into human behaviour. This framework outlines three core areas where we consider it to be strongest for application to nature and where this can go in the future for social science.



Thank you

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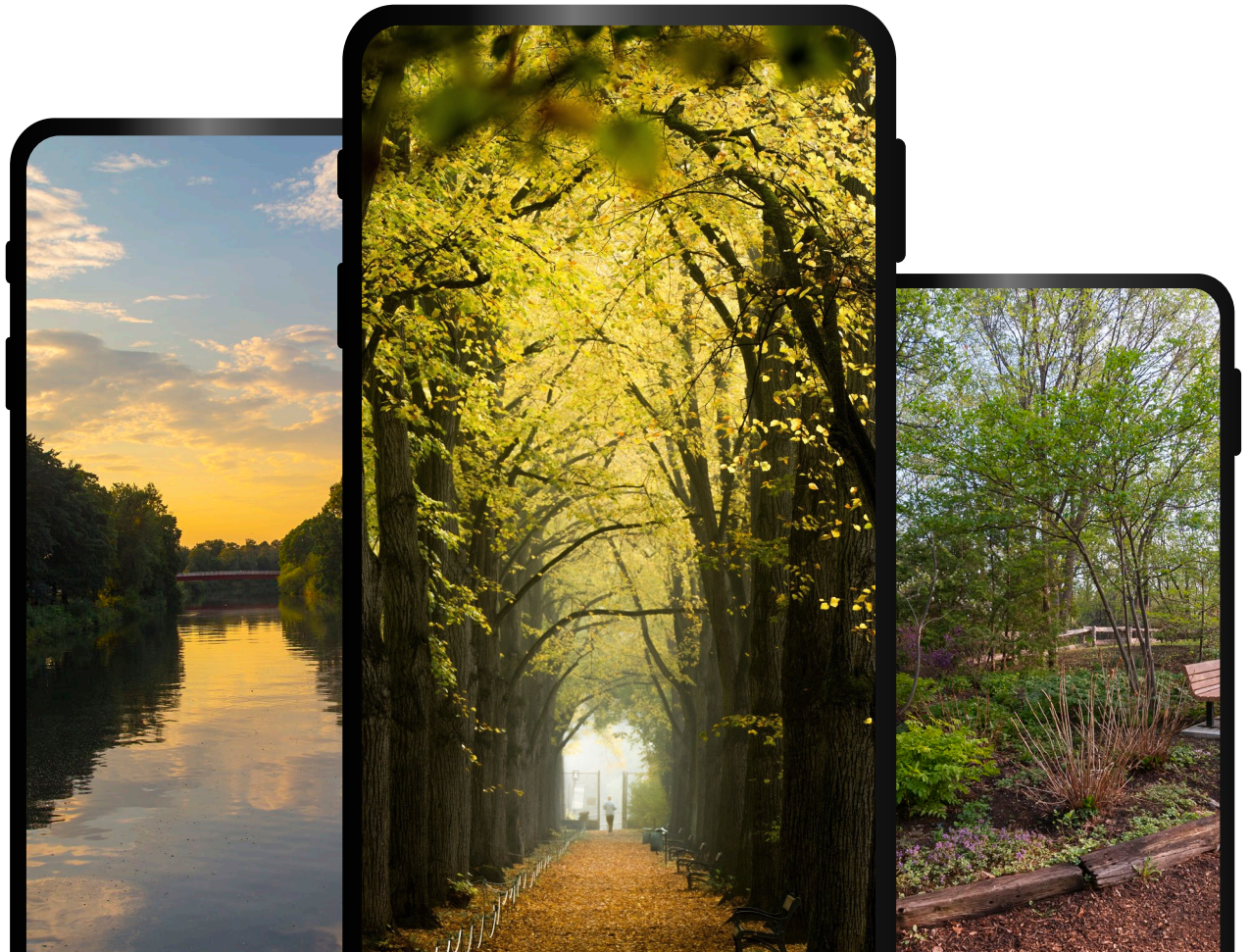
Full webinar:



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